

Level blockout: two Team Deathmatch arenas

Mohsin Raza Ojla · BSSE23040 · ITU

GitHub repository: github.com/bsse23040/ITU-CS-464-LAB-03-BSSE23040

Game I chose: PUBG Mobile, in its Team Deathmatch mode (the warehouse map).

For this lab I made two blockout levels in Unity using the provided POLYGON Prototype Pack by Synty together with Unity's cube game objects, then flew a camera through them in the Scene view and recorded it. Both levels are built for fast four-versus-four fights, because that is what I actually play in PUBG Mobile. This document has my reference images, screenshots of both levels, an explanation of each level and its obstacles, and why I designed them this way. The screen recording is a separate file, LevelWalkthrough.mp4.

1. Reference material

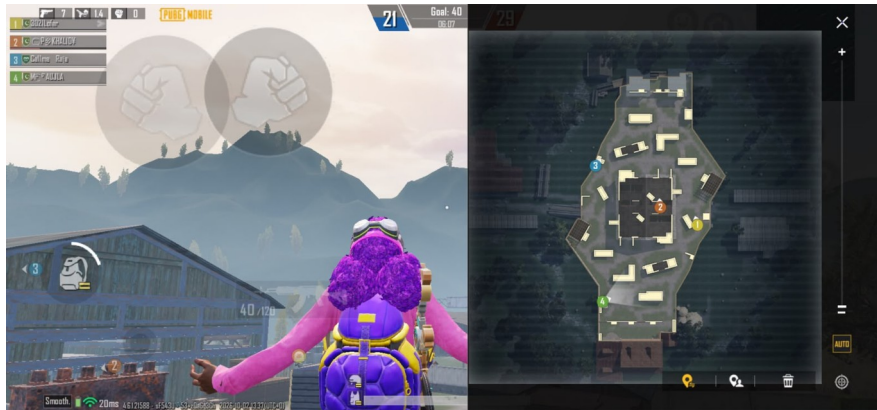
These are screenshots I took in a Team Deathmatch match, and I also recorded a short gameplay video (PUBG_gameplay_video.mp4, in the References folder of the repository). They are what I looked at while I built the levels.



Reference 1: a big two-storey warehouse in the middle, with containers and crates around it.



Reference 2: a lane between a wall of containers and wooden crates. An enemy is standing on a container roof.



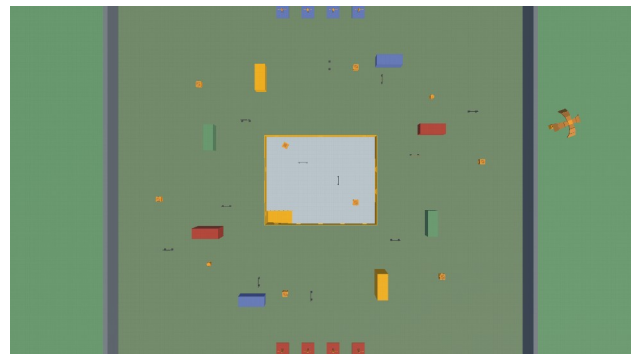
Reference 3: the full map. The arena is small, with a big central building, containers around it and spawns at both ends.

What I took from them: the arena is compact so teams meet within seconds. The middle is held by one large building. Shipping containers and crates are placed all over as cover, and some of them can be climbed, so height matters. Teams spawn at opposite ends, and every fight has to be won at a corner, a doorway or a roof edge.

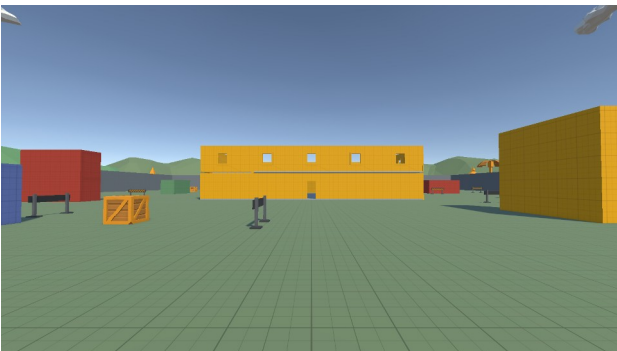
2. Level 1: Warehouse (Level1_Warehouse.unity)



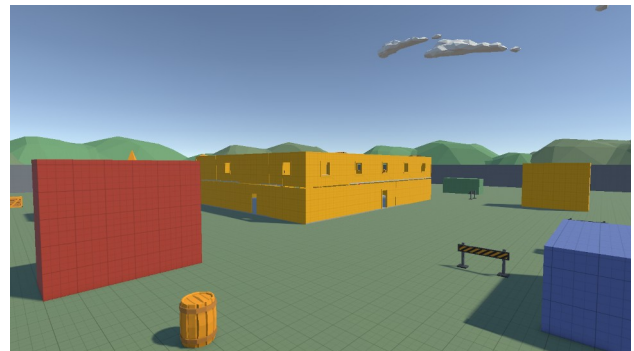
Overview: the whole arena from a corner, with Synty hills behind.



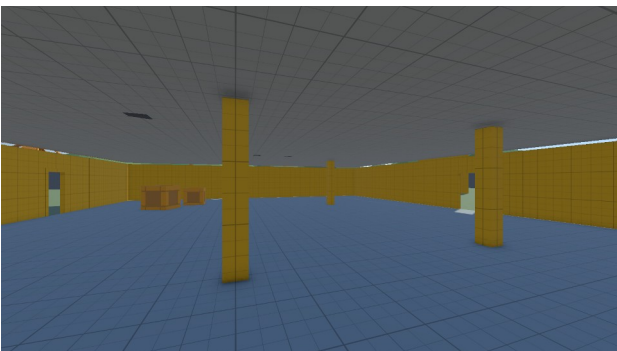
Top-down: red spawns at the bottom, blue at the top.



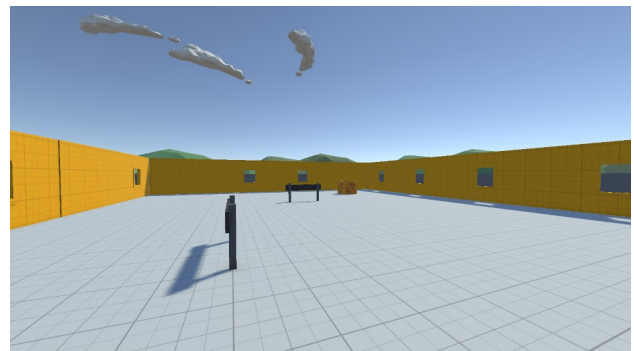
What a player sees from the red spawn.



The warehouse from the outside.



Ground floor, looking toward the pillars.



Upper floor, with window openings.

What the level is

A square arena of 96 m by 96 m inside a 5 m tall perimeter wall, with Synty hills, trees and clouds around it. The big landmark is a two-storey warehouse in the centre, 25 m by 20 m, built from the pack's 5 m wall modules. Both teams spawn on pads at their own end: four red pads on one side and four blue pads on the other, each with a Synty dummy character standing on it.

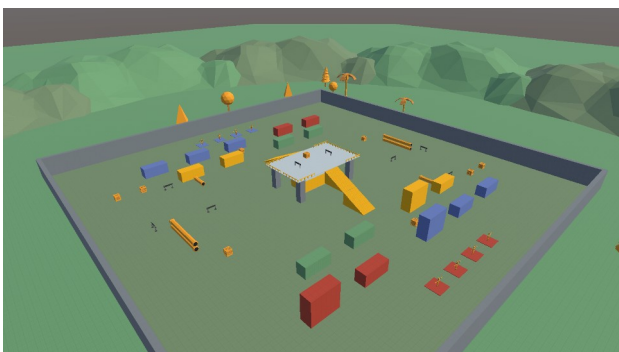
Obstacles and how they work

- **Warehouse doorways and windows.** The ground floor uses Synty WallDoor and Wall modules, with a doorway on every side, so a fight can come from four directions. The upper floor uses WallWindow modules for sniping openings.
- **Interior stairs.** The only way up is a 3 m wide stairway made of three Synty Stairs pieces side by side, climbing 3 m over 6 m in the south-west corner. Players on the stairs are exposed, and the rail around the stair opening upstairs lets people look down at them.
- **Columns and cover.** Four Synty columns on the ground floor and Synty barriers and crates on both floors break up the long sight lines inside.
- **Containers.** Ten shipping containers (6 m by 2.6 m by 2.4 m) are placed outside, two of them stacked. The pack has no container, so I made them from its scaled block piece in its red, blue, green and yellow grid materials. They block the long view across the arena and make lanes, like the container lane in my reference.
- **Scattered cover.** Synty crates, barriers, barrels and pipes (1.1 to 1.2 m high once scaled) are scattered in mirrored pairs across the open ground, so a player crossing the yard always has something to duck behind.

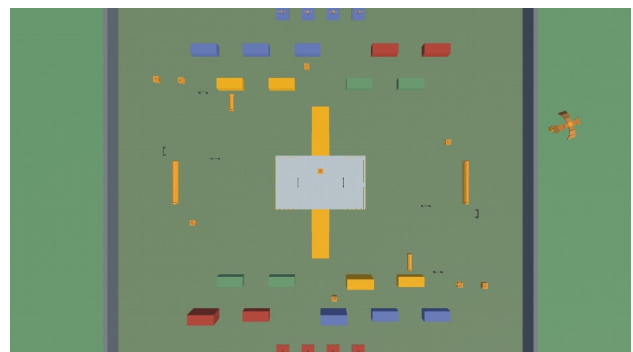
Why I designed it this way

- It copies the reference: one dominant building in the middle with containers around it, so the first thing both teams want is the middle.
- Everything is mirrored through the centre. A spawn on one side has exactly the same cover and distances as the other, so the level is fair.
- It gives three kinds of fights: long range across the yard, medium range between containers, and close range inside the building.
- Low cover is 1.1 to 1.2 m, high enough to crouch behind and low enough to shoot over when standing.

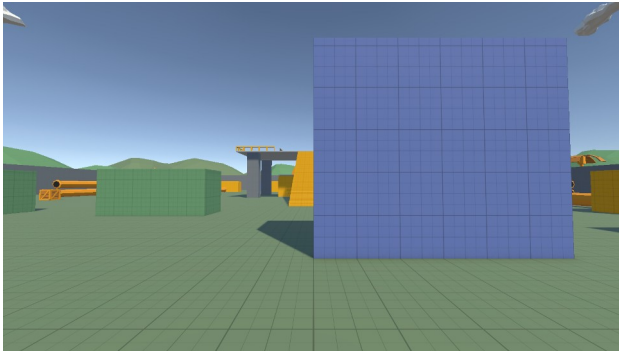
3. Level 2: Docks (Level2_Docks.unity)



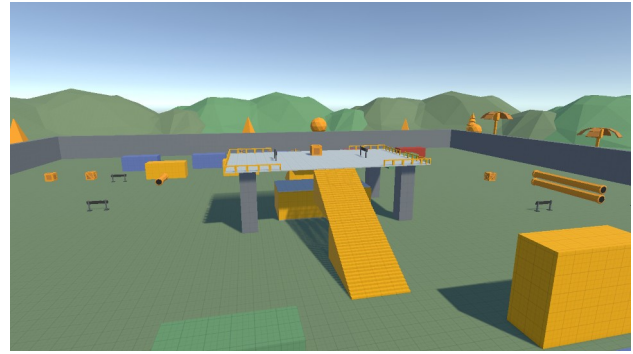
Overview: container lanes with the crane in the middle.



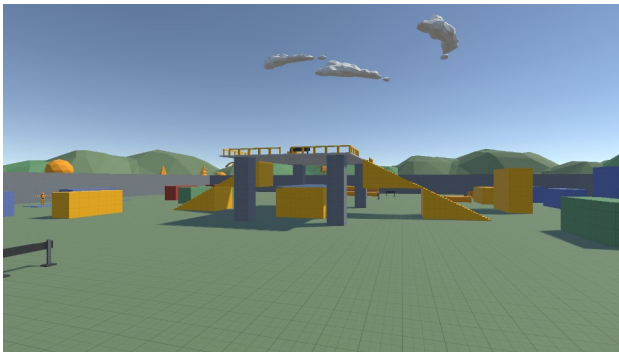
Top-down: two staggered rows of containers on each side.



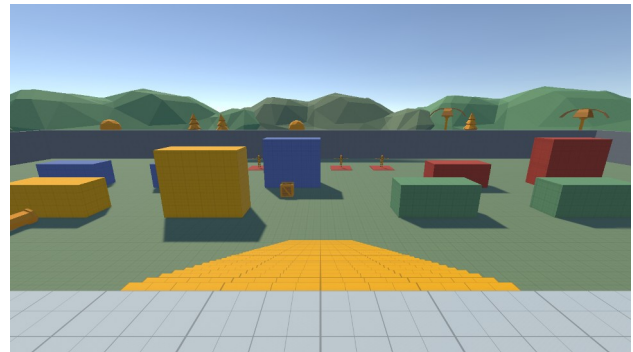
What a player sees from the red spawn.



The crane gantry and its stairs.



Side view: deck on four legs, stairs on both ends.



From the deck, looking over the containers.

What the level is

The same 96 m by 96 m arena and spawn layout, but a very different shape. Two staggered rows of containers fill each half, leaving lanes about 5.6 m wide. In the middle stands a crane gantry: a 20 m by 12 m deck on four legs, 6 m above the ground.

Obstacles and how they work

- **Container rows.** Nine containers per side, in two rows with gaps between them, made from the pack's scaled block piece. The rows are staggered so no lane is straight for the whole length, which stops a single player from holding it.
- **Long stairs.** Two long stairways, one from each team's side, lead up to the deck. Each is two flights of Synty Stairs (3 m rise per 6 m), so the climb is 6 m high over 12 m and completely exposed. Going up is a real risk.
- **Gantry deck.** The deck is the strongest sniping spot on the map. It has Synty rails, two barriers and a crate, and the rails are open where the stairs arrive.
- **Control hut.** A small hut under the deck (10 m by 5 m, one doorway, built from Synty wall modules) gives a close-range fight for players who stay on the ground.
- **Cover in the gaps.** Synty crates, barriers, barrels and long pipes fill the gaps between the container rows.

Why I designed it this way

- The first level is about one building and fights at doors. I wanted the second one to play differently, so it is about height: whoever holds the deck sees the whole arena.
- The long stairs balance the deck. It is strong, but a team that wants it has to climb while the other team shoots at them, so it needs teamwork.
- The container rows copy the lane I saw in my second reference screenshot, and the staggered gaps stop any lane from becoming a shooting gallery.
- Same spawn pads and the same mirrored layout as Level 1, so both levels stay fair to both teams.

4. How it was built

The lab asked for game objects and the provided assets, so I used both. The provided asset is the POLYGON Prototype Pack by Synty. I imported only the 196 assets my levels needed (about 7 MB) into Assets/Synty. From it I used the 5 m wall modules (plain, doorway and window), 5 m floor tiles, stairs, rails, columns, crates, barriers, barrels, pipes, a dummy character for each spawn point, and the hills, trees and clouds for the scenery. Cube game objects (the pack's scaled block piece) make the ground, the perimeter wall, the spawn pads, the container shapes and the crane legs, because the pack has no container.

A small editor script (Assets/Editor/SyntyLevels.cs) places every piece by its measured size, so the walls and floors line up on a 5 m grid. The levels are in Assets/Scenes/Lab03 in the repository. I used Unity's Scene view to look around, and the walkthrough video is a camera path flown through the Scene view of both levels.

	Level 1: Warehouse	Level 2: Docks
Main idea	One big building and doorway fights	Height, with a crane deck and container lanes
Strongest spot	The upper floor of the warehouse	The crane deck, 6 m up
Main cover	Containers, crates, columns	Container rows, barriers, pipes
Way up	One 3 m wide stairway inside	Two long stairways outside (6 m rise over 12 m)
Arena size	96 m by 96 m	96 m by 96 m

The Scene view walkthrough of both levels is in LevelWalkthrough.mp4, and everything (scenes, scripts, screenshots, references) is in the GitHub repository linked at the top.